

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

2026-2027 (Semester 1)

MA1513

Lecture Problem Set 2

Instructions

- To be discussed during lecture in week 2 (Aug 17 and 18, 2026).
 - Note that you do not need to submit the answers to this problem set.
 - To prepare for the problem set, students should view the "Week 2 key video" before the lecture.
 - Students who would like to try out the problems in advance may refer to the hints given at the end of the problem set.
 - You need to install MATLAB software and use the commands for some of the questions. (Refer to the PDF document: MATLAB for Engineering for details)
1. A manufacturer makes three types of chairs A, B, C. The company has available 260 units of wood, 60 units of upholstery and 240 units of labor. The manufacturer wants a production schedule that uses all of these resources. The various products require the following amounts of resources.

	A	B	C
Wood	4	4	3
Upholstery	0	1	2
Labor	2	4	5

- (a) Find the inverse of the data matrix above and hence determine how many pieces of each product should be manufactured.
- (b) If the amount of wood is increased by 10 units, how will this change the number of the three types of chairs produced? How are the answers related to the first column of the inverse matrix in (a)?
2. (a) Find the determinant of each of the following matrices.

$$(i) \mathbf{A} = \begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{pmatrix}, \quad (ii) \mathbf{B} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad (iii) \mathbf{C} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 4 \end{pmatrix}.$$

- (b) (**MATLAB**) Use MATLAB command **det** to check your answers in part (a).
- (c) Which matrices above are non-singular?

- (d) Without any further computation, can you tell whether $\mathbf{AB}$ and $\mathbf{CC}^T$ are singular?
- (e) Without performing Gaussian Elimination, can you tell whether the homogeneous systems $\mathbf{ABx} = \mathbf{0}$ and $\mathbf{CC}^T\mathbf{x} = \mathbf{0}$ have non-trivial solutions? Why?
3. Let $\mathbf{u}_1 = (2, 1, 0, 3)$, $\mathbf{u}_2 = (3, -1, 5, 2)$, and $\mathbf{u}_3 = (-1, 0, 2, 1)$ and $\mathbf{v} = (2, 3, -7, 3)$, $\mathbf{w} = (1, 1, 1, 1)$.
- (a) Are $\mathbf{v}$ and $\mathbf{w}$ linear combinations of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$?
- (b) Do $\mathbf{v}$ and $\mathbf{w}$ belong to $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$?
- (c) Is $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{v}\}$?
- (d) Is $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \mathbb{R}^4$?
- Justify your answers.
4. (a) How are $\text{span}\{(1, 0, 0)\}$, $\text{span}\{(-1, 0, 0)\}$ and $\text{span}\{(1, 0, 0), (-1, 0, 0)\}$ related? Describe what geometrical objects do they represent in the xyz -space.
- (b) Recall that the span of two non parallel vectors in $\mathbb{R}^3$ represents a plane in the xyz -space that contain the origin.
- (i) Find the linear equation of the plane P that represents $\text{span}\{(2, -1, 0), (-1, 1, 1)\}$.
- (ii) Show that the vector $(-2, -4, 2)$ is orthogonal to both $(2, -1, 0)$ and $(-1, 1, 1)$. Use this to explain why vector $(-2, -4, 2)$ is perpendicular to all vectors on the plane P in (i).

Hints

1. (a) $\begin{pmatrix} \frac{3}{2} & 4 & -\frac{5}{2} \\ -2 & -7 & 4 \\ 1 & 4 & -2 \end{pmatrix}.$

Let x, y, z be the number of type A, B, C chairs to be manufactured respectively. Then set up a linear system in matrix equation form using the data matrix provided. Then solve for x, y, z using the inverse matrix approach. 30 chairs of type A, 20 chairs of type B and 20 chairs of type C.

(b) A increased by 15; B decreased by 20; C increased by 10.

2. (a) 0, 6, 24 (Compute the determinants directly by expanding into smaller determinants).

(c) There is a relation between determinant and singularity of matrices.

(d) There is a relation between singularity of matrices and the solutions of the corresponding homogeneous systems.

3. (a) For each of $\mathbf{v}$ and $\mathbf{w}$, you need to set up a vector equation and convert into a linear system.

(b) The answers follow from those of (a) immediately.

(c) The answer follow from that of (b).

(d) No.

4. Refer to chapter 2 notes, last part of section 2.2.

For (b) part (i), recall from section 1.1 that a plane in xyz -space is represented by a linear equation of the form $ax + by + cz = d$. (Ans: $x + 2y - z = 0$.)

For part (ii), use dot product. Note that every vector on the plane P is a linear combination of $(2, -1, 0)$ and $(-1, 1, 1)$.